THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A convex mirror is used as a rear-view mirror. Reason (R): It always forms virtual, erect and diminished images with a wide field of view. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): The image formed on the retina is inverted. Reason (R): The brain interprets the signals to make the object appear erect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): A voltmeter is connected in parallel with a resistor. Reason (R): A voltmeter has a very high resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Magnetic field lines never intersect. Reason (R): The field at a point has only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A concave mirror forms a real image of a candle on a screen kept at 45 cm. If the candle is 15 cm from the mirror, the focal length of the mirror (magnitude) is 1

(a) 11.25 cm **(b)** 30 cm **(c)** 22.5 cm **(d)** 15 cm

6. A person with a far point at 25 cm requires spectacles to see distant objects clearly. The focal length of the lens is 1

(a) +25 cm **(b)** -25 cm **(c)** -31 cm **(d)** -6 m

7. The resistance of a wire of length L and area A is 12 ohm. A second wire of the same material has length 2L and radius double that of the first. Its resistance is 1

(a) 24 ohm **(b)** 12 ohm **(c)** 3 ohm **(d)** 6 ohm

8. A straight wire with current is placed between the poles of a magnet. The force on the wire is greatest when the wire makes an angle with the field of 1

(a) 90 degrees **(b)** 45 degrees **(c)** 30 degrees **(d)** 0 degrees

9. An object is placed 24 cm from a lens and a real image forms at 24 cm. The lens is 1

(a) convex, f = 12 cm **(b)** convex, f = 24 cm

(c) concave, f = -12 cm **(d)** concave, f = -24 cm

10. The sky is blue because 1

(a) larger wavelengths are scattered more by small particles

(b) smaller wavelengths are scattered more by small particles

(c) the sun emits blue only

(d) the sea reflects blue

11. Which graph is a straight line through the origin for a metallic conductor at constant temperature? 1

(a) V against I **(b)** P against I **(c)** I against R **(d)** V against t

12. A generator coil of frequency 50 Hz is rotated at double the speed. The frequency becomes 1

(a) 50 Hz **(b)** 25 Hz **(c)** 200 Hz **(d)** 100 Hz

13. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles. 1

Reason (R): A convex mirror always forms an erect, diminished image and gives a wider field of view.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** Myopia is corrected by using a concave lens. 1

Reason (R): A concave lens diverges the rays so that the image is formed farther back, on the retina.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** An ammeter is connected in series in a circuit. 1

Reason (R): An ammeter has a very low resistance.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The image of an object formed by a concave mirror is real, inverted and 1/3 of the object size at 15 cm from the mirror. Find the object distance and focal length. 2

17. Why does the amount of scattering of red light not matter much in the colour of the sky? 2

18. Calculate the number of electrons flowing per second in a wire carrying 4.8 mA. 2

19. A charged particle enters a magnetic field and its speed remains unchanged. Explain why. 2

20. A student uses a concave mirror to focus the Sun's rays on paper kept at 20 cm. Find the radius of curvature of the mirror. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A concave mirror forms a real image three times larger than the object. When the object is moved 10 cm farther away, the image becomes the same size as the object. Find f. 3

22. A person with a near point at 30 cm wants to read at 25 cm. Find the power of the lens required. 3

23. The resistance of a copper wire is 2.5 ohm at 20 degrees Celsius. Why does its resistance rise at high temperature, and what is the effect on the current if the voltage is constant? 3

24. Explain why the force on a current-carrying conductor depends on the angle between the conductor and the field. 3

OR

A student wants a 5 times magnified real image on a screen using a convex lens of f = 12 cm. Find the object distance and the screen distance. 3

25. Why does the sky appear whitish near the horizon on a clear day? 3

OR

Three bulbs of 100 W each rated 220 V are connected in parallel with a 220 V source. Find the total current and the equivalent resistance. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A student places a 3 cm tall object on the principal axis of a concave mirror of radius of curvature 40 cm, at a distance of 30 cm from the pole. 5
(i) Find the focal length of the mirror.
(ii) Find the position of the image.
(iii) Find the magnification.
(iv) Find the height of the image and state its nature.

27. A cell of 12 V and negligible internal resistance is connected to a 6 ohm resistor in series with a parallel pair of 12 ohm and 4 ohm. Find 5
(a) the equivalent resistance
(b) the main current
(c) the current through each resistor in parallel
(d) the power consumed in the circuit.

OR

A straight wire carries current from west to east in a field pointing north. Find the direction of the force and describe how it changes 5
(a) when the current is reversed
(b) when both are reversed. State the rule used.

END OF QUESTION PAPER Total: 50 Marks